The paper I have selected :

Howard MA, Geist MA, Ordemann GJ, et al. Behavioral and Prefrontal Circuit Deficits in a Newly Developed Setbp1 Haploinsufficiency Mouse Model. *Biol Psychiatry Glob Open Sci*. 2026;6(2):100666. doi:10.1016/j.bpsgos.2025.100666

Test 1:

In the methods section the authors state that datasets are underpowered when divided by sex, but that this is included for insights into trends in the data. In the results of the open field test the authors state “When sex was included as a variable, 2-way ANOVA revealed a significant effect of genotype but not of sex and no significant interaction between sex and genotype (2-way ANOVA; main effect of genotype: $p = .0007$; main effect of sex and sex × genotype interaction: $p > .05$)”, and do not revisit this point again.

By concluding that there is no significant interaction they are accepting the null hypothesis, despite the fact they acknowledge that this analysis is underpowered. This undermines their point.

I think that a more appropriate restatement would say “no sex or sex x genotype interactions were clearly identified at the available sample size”.

I think it is particularly important to not use such conclusive language here because upon visual analysis of the distance travelled by group when stratified by sex, you can see that the Setbp1 mutant females are highly variable in their behaviour, which can impact p-value, but doesn’t mean that there isn’t a relationship between sex and genotype (Figure 1C).

Test 2:

When describing the results of a movement assay the authors stated that “Despite increases in distance traversed, there was no difference between genotypes in the amount of time spent in the center zone (Mann-Whitney test, $p > .05$) ”. This is not true, the groups spent similar but distinctly different amounts of time in the centre zone. What they mean to say is that they didn’t find a p>0.05.

Later in that same paragraph the authors say that the genotypes show “equivalent phenotypes” which I think is a more appropriate representation of their findings. Equivalent phenotypes suggests that the differences are minimal enough that they are choosing to ignore them for future analyses. This is much clearer than saying that there are no differences between groups (which is not true).

I would revise the initial statement to say that “despite increases in distance traversed, both genotypes spent less than a third of the assay in the centre zone, displaying equivalent phenotypes”.

Test 3:

When comparing neuron excitability the authors stated “These data show that above the near-threshold range, *Setbp1*$^{+/-}$ neurons spiked a similar number of times compared with WT neurons (2-way RM ANOVA, effects of genotype and genotype × current interaction: $p > .05$)”.

I think that their use of the term similar was appropriate. It doesn’t accept the null hypothesis or make any big conclusions about their neuron function being the same because of this.

One thing that I think would have been a good addition to this statement would be including the actual p-value in text, rather than just p>.05. If the p-value is something that holds a lot of weight to the author, I think the most appropriate thing is to include it in text. As a reader, generalizing p-values of 0.06 and 0.3 is an overgeneralization of the statistics, and is following the “golden rule” of 0.05 too closely. This may be obstructing a biologically relevant trend. Thus, for clarity and to allow the reader to critically think about the results, the p-value should be included.